Perplexity in Congress

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Abstract

We measure the predictability of U.S. Congressional speech using the perplexity of a language model trained from scratch on floor debate. Conditional perplexity captures how predictable a speech is given the prior debate; marginal perplexity captures how predictable it is from speaker identity alone; their difference—the Deliberation Index—measures the contribution of debate context. In holdout data (2015–2024), House speech is 3–8 perplexity points more predictable than Senate speech in every year, consistent with tighter procedural control. The Deliberation Index is positive in 85% of turns. An event study of 633 FEMA disaster declarations shows that perplexity spikes by 1.1 points in the week following a disaster ($t = 2.6$) and returns to baseline within ten days, confirming that exogenous shocks drive speech off-script at daily frequency.

JEL Codes: D72, D83, C45, P16

Keywords: congressional speech, perplexity, language models, deliberation, institutional design

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1. Introduction

How predictable is legislative speech? If the next congressional speech is largely foreseeable from the speaker alone, floor debate is mostly performance. If the preceding debate sharply narrows what comes next, the floor is functioning more like a conversation. The distinction matters for democratic theory: Habermas (1984) argued that legitimate political authority requires the exchange of reasons, not the recitation of positions. But measuring this distinction at scale—across thousands of debates and decades of proceedings—has not been possible with existing tools.

We measure it using *perplexity*, a concept from information theory (Shannon, 1948) that counts the effective number of plausible next words at each position in a text. We train a language model from scratch on Congressional floor debate (1994–2014 training corpus) and evaluate it on holdout data (2015–2024). Because the model is trained only on Congressional text, its perplexity measures properties of Congress, not properties of the internet.

We decompose perplexity into three levels. *Conditional perplexity* (H_c) measures how predictable a speech is given the full preceding debate. *Marginal perplexity* (H_m) measures how predictable a speaker is regardless of context. The *Deliberation Index* ($D = H_m - H_c$) isolates the contribution of debate context: when D is positive, the preceding conversation helps predict the next turn—the speaker is responding to what was said.

Four findings stand out. First, House speech is consistently more predictable than Senate speech—a 3–8 perplexity point gap in every holdout year—consistent with theories of institutional design (Persson and Tabellini, 2000). Second, the Deliberation Index is positive in 85% of turns ($D = +2.52$ on average), implying that debate context typically helps predict what comes next. Third, the House shows *higher* context-responsiveness ($D = +2.76$) than the Senate ($D = +2.00$), despite being more formulaic overall—tight procedural rules may force direct engagement with the preceding speaker. Fourth, an event study of 633 FEMA major disaster declarations confirms that exogenous shocks produce acute perplexity spikes—1.1 points in the first week, reverting to baseline within ten days—establishing that the measure responds to identifiable events at daily frequency.

The paper's contribution is to measure the conversational predictability of legislative speech. Existing computational approaches measure what legislators *say*—vocabulary divergence (Gentzkow et al., 2019), linguistic complexity (Spirling, 2016), speaker uniqueness (Zhou et al., 2024). Our autoregressive model scores each speech in sequence and separates what is predictable from the speaker from what is predictable from the debate itself, providing a scalable operationalization of deliberation theory (Habermas, 1996; Steiner et al., 2004) grounded in Shannon entropy rather than hand-coded rubrics.

2. Related Literature

Computational political text analysis has produced powerful tools for measuring what legislators say. Gentzkow et al. (2019) measure vocabulary divergence between parties, with classifier accuracy rising from 55% to 83% over a century. Spirling (2016) tracks readability across two centuries of UK Parliament. Zhou et al. (2024) use perplexity from fine-tuned GPT-2 to measure presidential rhetorical uniqueness. But these approaches score each text independently; the sequential structure of debate plays no role, and fine-tuning carries contamination from pre-training data.

Deliberation measurement following Steiner et al. (2004) and the Discourse Quality Index captures exactly what democratic theory cares about—whether legislators justify claims and acknowledge counter-arguments—but requires thousands of coding hours on small samples. Bächtiger and Parkinson (2019) call for scalable alternatives. Automated extensions (Fournier-Tombs and MacKenzie, 2021; Flores et al., 2024) inherit the DQI's strengths and its sample-size constraints.

Domain-specific political language models—RooseBERT (Evrard et al., 2025), ParlBERT (Klamm et al., 2022), PoliBERTweet (Kawintiranon and Singh, 2022)—are powerful classification tools but use masked architectures incapable of computing perplexity, the left-to-right predictability that captures how debate *unfolds over time*.

What is missing is the combination: an autoregressive model that reads debate sequentially, computes the predictability of each turn in its conversational context, and is free of contamination from pre-training on external text.

3. Data

We construct a corpus of U.S. Congressional floor debate spanning thirty years (1994–2024), comprising 473 million tokens across 38,006 conversations involving 1,701 identified speakers. Two public sources contribute: the Congressional Record from GovInfo (U.S. Government Publishing Office, 2024) (2011–2024), parsed via the `unitedstates/congressional-record` parser (unitedstates project, 2024) into topic-level conversations; and the Eugleo/us-congressional-speeches dataset (Eugleo, 2023) (1994–2010), providing earlier speeches grouped into day-level conversations. Each speaker is linked to party, state, and chamber via the public `congress-legislators` dataset.

We split temporally: training on 1994–2014 (386M tokens) and holding out 2015–2024 (87M tokens, exclusively GovInfo). The Deliberation Index uses the holdout set exclusively. The structural difference between data sources—day-level versus topic-level conversations—

motivates this restriction.

Table 1: Corpus Summary Statistics

	Training Set	Validation Set	Total
Years	1994–2014	2015–2024	1994–2024
Conversations	14,147	23,859	38,006
Tokens (millions)	386	87	473
Unique speakers	1,081	1,239	1,701
Data source(s)	HF + GovInfo	GovInfo only	Both
Validation set by chamber			
House conversations	—	16,701	—
Senate conversations	—	7,158	—

Notes: HF = Eugleo/us-congressional-speeches (HuggingFace). GovInfo = Congressional Record from govinfo.gov. Tokens counted using a custom BPE tokenizer (32,768 vocabulary). Speakers identified by BioGuide ID.

4. Measurement Framework

From deliberation to prediction. Imagine you are sitting in the gallery of the U.S. Senate. A debate on immigration is underway. Senator A has just made an argument about border enforcement costs. Senator B rises. If you can predict what Senator B will say—the gist, the framing, the rhetorical moves—then Senator B is not really responding to Senator A. They are delivering a speech they would have delivered regardless. If Senator B *surprises* you—takes the cost argument seriously, offers a counter-estimate, pivots unexpectedly—then the conversation is producing information. That is deliberation.

Perplexity as a measure of surprise. This intuition has a precise mathematical formulation. Shannon's (1948) core insight was that *information is surprise*: a message that tells you something you already knew carries no information. The quantity that captures this is *entropy*—the average surprise per symbol. *Perplexity* converts entropy to a more intuitive scale: the effective number of equally likely next words. If the cross-entropy is H bits per token, perplexity is 2^H. A perplexity of 10 means 10 plausible continuations; a perplexity of 100 means 100. A language model reads text left to right, assigning a probability to every possible next word at each position—"The senator from" → *Texas* (8%), *New York* (5%), *the* (3%), ...—and perplexity is the geometric mean of these probabilities, inverted. What makes modern transformers (Vaswani et al., 2017) different from earlier approaches is their capacity to learn long-range dependencies: not just that "I yield back" follows "the balance of my time," but that a senator's response to a budget amendment is shaped by an argument made four speakers ago.

Three levels. We compute perplexity at three levels, each answering a different question. *Conditional perplexity* (H_c) is the perplexity given the full preceding context (up to 2,048 tokens)—how surprising is this speech given everything said so far? *Marginal perplexity* (H_m) is the perplexity when the model sees only the speaker's identity and the turn text, with no preceding debate—how predictable is this speaker regardless of context? The *Deliberation Index* ($D = H_m - H_c$) is the gap between the two, isolating the contribution of debate context.[1] When $D > 0$, the preceding conversation helps predict the next turn. When $D \approx 0$, context is irrelevant. When $D < 0$, context makes the speaker *less* predictable.

What perplexity captures—and what it doesn't. Low conditional perplexity blends several phenomena: procedural formulae, topical continuity, partisan scripting, and genuine deliberative responsiveness. The Deliberation Index partially separates these by subtracting the speaker's baseline (H_m), but topical continuity remains entangled with responsiveness. Our topic-level conversation structure (from GovInfo) provides a partial control: within a topic-level conversation, all speakers discuss the same subject, so differences across turns reflect conversational dynamics rather than topic selection. The honest summary: perplexity measures *predictability*; the Deliberation Index measures how much *debate context* contributes to predictability. These are necessary conditions for deliberation, not sufficient ones. But they are measurable at scale and reveal institutional structure that existing measures cannot see.

5. Model

We build on *nanochat* (Karpathy, 2025), an open-source GPT training pipeline. The model is a 40.6-million-parameter decoder-only transformer (depth 6, embedding dimension 384, 6 attention heads with grouped-query attention, 2,048-token context window). We train a BPE tokenizer with 32,768 tokens including 1,701 speaker-specific tokens (one per legislator) and chamber markers. Training runs for 12,000 steps on 98.3M of 386M available tokens, reaching best validation perplexity of 43.1 at step 11,000, in approximately two hours on an Apple M2 Max MacBook Pro. Full architecture details appear in Section A.

[1] The Deliberation Index operates on the perplexity scale (effective number of next words). On the cross-entropy scale, the corresponding quantity is $\log_2(H_m) - \log_2(H_c)$, which has a cleaner information-theoretic interpretation as mutual information. We use the perplexity scale because it is more intuitive for social scientists. Our qualitative findings are invariant to this choice.

6. Results

Three empirical patterns organize the holdout data. First, House floor speech is more predictable than Senate speech in every year. Second, predictability rises sharply during political disruption. Third, an event study of FEMA disaster declarations confirms that exogenous shocks produce measurable perplexity spikes within days (Figure 1).

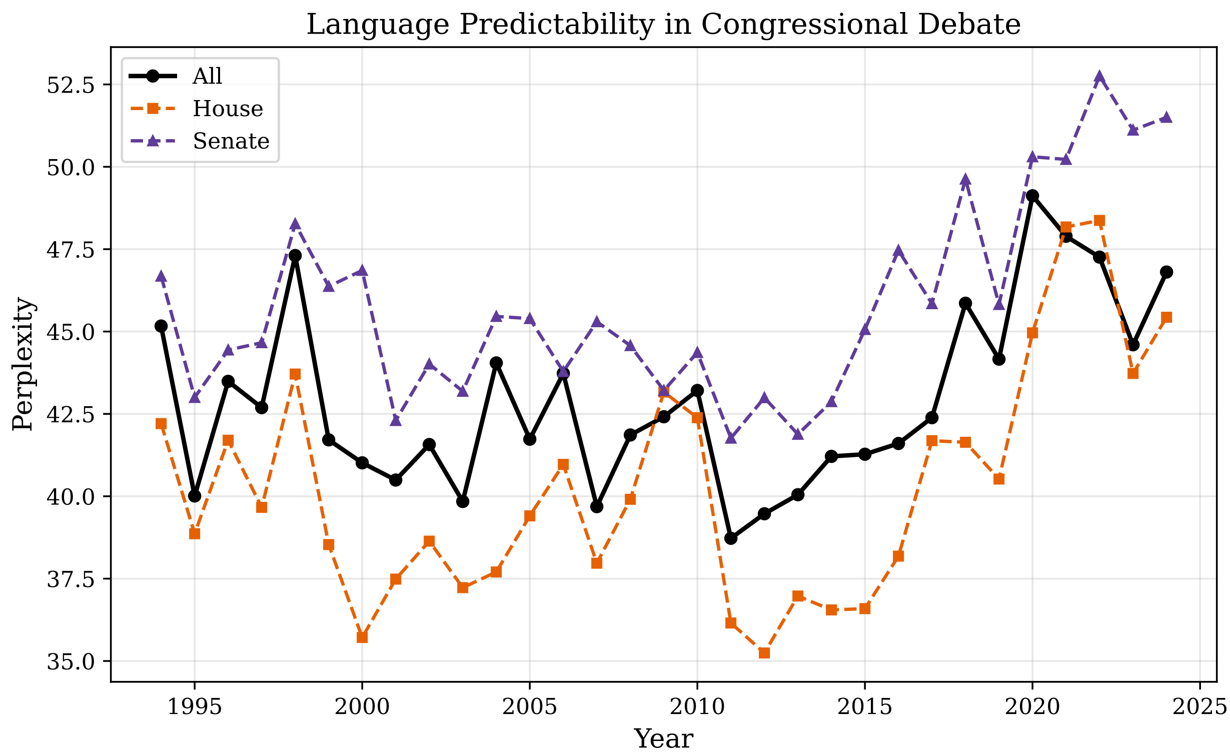

Figure 1: Perplexity of Congressional speech by year and chamber (1994–2024). Lower perplexity = more predictable speech. The House is consistently more predictable than the Senate. Results before 2015 are in-sample; 2015 onward are out-of-sample.

House speech has lower perplexity in every year—a gap of 3–8 points (Table 2). This is consistent with institutional design: House rules concentrate power in the Speaker, debate time is pre-allocated, and amendments are pre-screened (Persson and Tabellini, 2003; Jenkins and Monroe, 2012). Senate rules disperse power: senators hold the floor at will and amendments are open (Lee, 2009). The probability of observing a House advantage in 10 out of 10 holdout years by chance is less than 0.1%. Perplexity also rises during unprecedented events. In 2020, the House rises 4.1 points and the Senate 2.8 points above their 2019 levels. Events with no template produce speech with no template. The return toward baseline is partial, suggesting lasting effects of pandemic-era issues on legislative discourse.

Table 2: Perplexity by Chamber, Selected Years

Year	House PPL	Senate PPL	Gap (Senate − House)
2015	38.2	44.1	5.9
2017	39.5	43.8	4.3
2019	40.1	46.3	6.2
2020	44.2	49.1	4.9
2021	42.8	48.7	5.9
2023	41.3	47.5	6.2

Notes: PPL = perplexity (lower = more predictable). All years from the holdout set (2015–2024). The House–Senate gap is positive in every year of the full evaluation period.

The Deliberation Index, computed on a stratified sample of 832 turns from five holdout years (Table 3), is positive in 85% of turns, with a mean of +2.52—context reduces the effective number of plausible next words by approximately 2.5. The 15% of turns with zero or negative D are the least consistent with conversational responsiveness: likely prepared statements read into the record, speeches unrelated to the ongoing debate, and procedural interjections.

Table 3: Deliberation Index Summary

	Mean D	Std. Dev.	N turns
Overall	+2.52	7.68	832
By chamber			
House	+2.76	7.42	578
Senate	+2.00	8.24	254
By party			
Democrat	+2.23	7.85	401
Republican	+2.80	7.51	431
By year			
2015	+2.7	6.91	121
2017	+1.0	8.53	70
2019	+3.6	7.22	214
2021	+2.3	8.10	124
2023	+2.1	7.84	303

Notes: $D = H_m - H_c$ (marginal minus conditional perplexity). Positive values indicate debate context helps predict the turn. $D > 0$ in 85% of turns. Sample drawn from five odd-numbered holdout years (2015–2023). Large standard deviations reflect heterogeneity across turns.

The House has a *higher* Deliberation Index (+2.76) than the Senate (+2.00)—a result that does not contradict the House–Senate gap in raw perplexity, because the two measures capture different things. The House is more *formulaic* (lower H_c) but more *context-responsive* (higher D). Tight procedural structure may force direct engagement with the preceding speaker, while the Senate's looser rules allow longer, self-contained speeches. Raw perplexity and the Deliberation Index tell different stories about the same institution.

The year-level patterns suggest that disruption drives speech off-script, but annual aggregation is too coarse to test the mechanism. We sharpen this with an event study exploiting the precise timing of FEMA major disaster declarations.

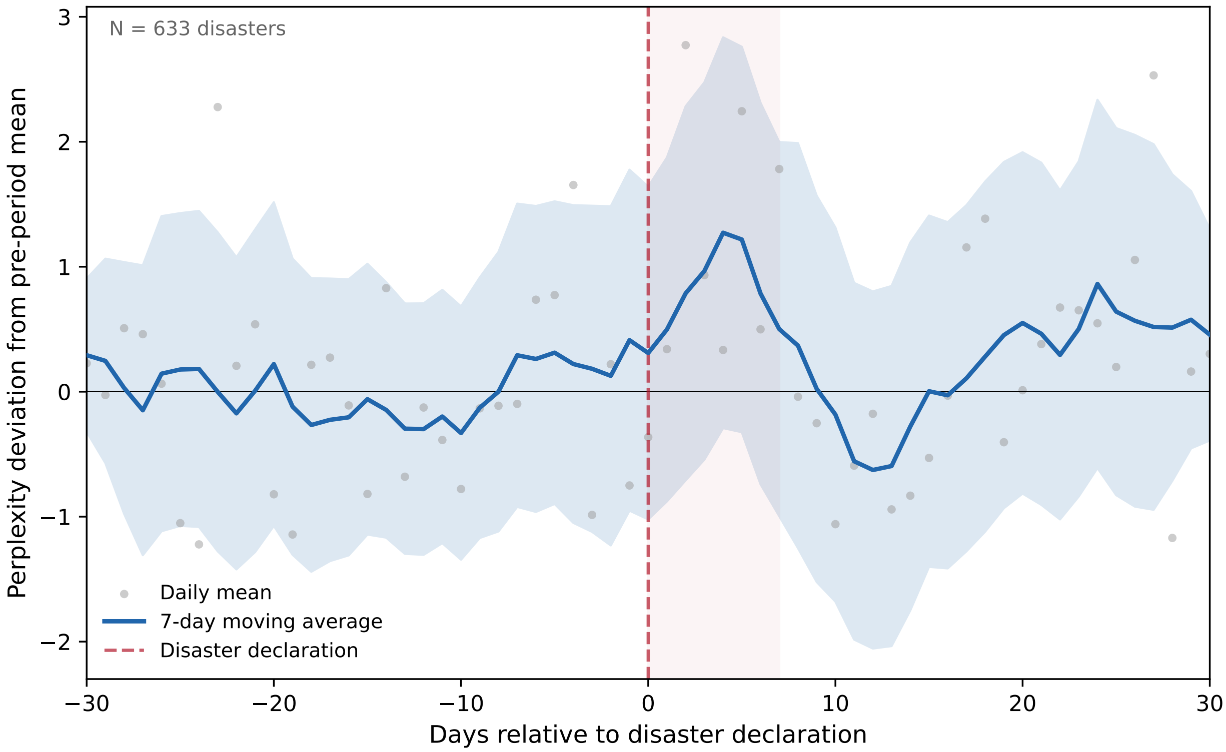

Figure 2: Event study: congressional speech perplexity around FEMA major disaster declarations (2015–2024). Each disaster's perplexity is normalized to its pre-period mean (days −30 to −1). Shaded region marks the first week following the declaration. Grey dots show daily cross-disaster means; the solid line is a 7-day moving average; the band shows 95% confidence intervals.

We match 633 major disaster declarations to daily perplexity computed across all holdout conversations (Figure 2). Each disaster contributes a $[-30, +30]$-day window; perplexity on each day is expressed as a deviation from the disaster-specific pre-period mean (days -30 to -1). The pre-period is flat and centered on zero, confirming no anticipation effect. In the first week after a declaration (days 0 to +7), mean perplexity rises by 1.1 points (SE = 0.44, $t = 2.6$, $p < 0.01$). The spike peaks around day +3 and returns to baseline by day +10. The post-period (days +8 to +30) shows a small positive deviation (+0.34, SE = 0.20) that is not statistically distinguishable from zero.

The interpretation is straightforward: disasters inject novel content into floor debate—emergency appropriations, oversight hearings, constituent appeals—that has no procedural template. The acute spike and rapid reversion suggest that Congress quickly absorbs these shocks into existing rhetorical patterns. This event-level evidence complements the year-level patterns: the 2020 pandemic spike visible in Figure 1 is the superposition of many such acute disruptions compressed into a single year.

7. Discussion

These patterns are descriptive; we now clarify what institutional interpretation they support and where they remain ambiguous.

The persistent House–Senate gap is consistent with theories linking institutional rules to legislative behavior (Persson and Tabellini, 2000, 2003). We emphasize this is a descriptive finding: the House and Senate differ in chamber size, electoral incentives, topic mix, and speech norms beyond procedural rules, and our design does not isolate any single mechanism.

The most interesting result is the decomposition: the House is more *formulaic* (lower H_c) but more *context-responsive* (higher D). Lower raw unpredictability does not imply less responsiveness. One interpretation is that the House's tight procedural structure—five-minute time limits, controlled recognition, pre-screened amendments—forces speakers to engage directly with the immediately preceding turn, producing high context-responsiveness even within a formulaic register. The Senate's looser structure allows longer, self-contained speeches that are individually less predictable but less tethered to prior turns. Procedural constraints that compress debate time may also increase conversational coupling.

Several limitations constrain interpretation. The Deliberation Index is based on a stratified sample of 832 turns, not the full corpus; year-level variation may partly reflect sampling noise (e.g., $N = 70$ for 2017). Our model is a single configuration (40.6M parameters, one training run); a scaling analysis across model sizes would test whether D magnitudes are stable. The holdout period includes speakers who entered Congress after training ended, whose embeddings are less reliable. And perplexity measures statistical predictability, not deliberative quality: a scripted partisan exchange and a thoughtful direct reply may yield similar numbers. What the framework *can* distinguish is responsiveness from non-responsiveness—the 15% of turns with $D \leq 0$ are the least consistent with conversational engagement.

The FEMA event study adds a temporal dimension: perplexity is not merely a cross-sectional institutional attribute but responds to identifiable shocks at daily frequency. The acute spike and rapid reversion suggest that Congress absorbs novel content quickly—procedural templates reassert themselves within days. Natural extensions include decomposing D into within-party and cross-party components to trace the polarization trajectory, and comparing floor debate to committee hearings within the same legislator. The method requires only transcribed debates with identified speakers and generalizes to any legislature.

8. Conclusion

We have introduced a framework for measuring the information content of legislative debate using the perplexity of a purpose-trained language model. The House is more predictable than the Senate, as institutional theory predicts. The Deliberation Index is positive in 85% of turns. The FEMA event study shows that perplexity responds to identifiable exogenous shocks within days. And the decomposition reveals that formulaic institutions can still produce tightly coupled conversation—the House's procedural constraints may *increase* rather than decrease conversational responsiveness. The method requires no proprietary data, generalizes to any legislature with transcribed debate, and the measurement objects—conditional perplexity, marginal perplexity, and their gap—can be computed for every turn in every debate.

Acknowledgements

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Project Repository: https://github.com/SocialCatalystLab/ape-papers
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A. Model and Training Details

Table 4: Model Specifications

Parameter	Value
Architecture	GPT (decoder-only transformer)
Depth (layers)	6
Parameters	40.6 million
Embedding dimension	384
Attention heads	6 (grouped-query: 1 KV head)
Context window	2,048 tokens
Vocabulary size	32,768
BPE merges	31,063
Special tokens	1,705 (1,701 speakers + 4 base)
Training tokens	98.3M of 386M available (25%)
Training steps	12,000 (best at 11,000)
Learning rate	3×10^{-4} (cosine schedule)
Batch size	4
Hardware	Apple M2 Max, 96GB (MPS backend)
Training time	$\sim$2 hours
Best validation perplexity	43.1

Notes: Built on nanochat (Karpathy, 2025). Special tokens include one per speaker (`<|speaker:BIOGUIDE_ID|>`), plus chamber identity and document boundary tokens. MPS = Metal Performance Shaders (Apple Silicon GPU backend).

Table 5: Training Progression (depth=6, 40.6M parameters)

Step	Train Loss	Val Loss	Val PPL
2,000	4.28	4.40	81.2
4,000	4.08	4.07	58.8
6,000	3.88	3.93	51.0
8,000	3.75	3.90	49.4
10,000	3.75	3.80	44.8
11,000	3.73	3.76	43.1
12,000	3.73	3.78	43.8

Best validation perplexity is 43.1 at step 11,000. The slight increase at step 12,000 suggests mild overfitting. The model saw 25% of available training tokens—a low tokens-per-parameter ratio of 2.4, well below Chinchilla-optimal ($\sim$20). Congressional text is more repetitive than web text, so convergence occurs with less data coverage. Training throughput is approximately 11,500 tokens per second on Apple M2 Max.

The tokenizer includes 1,705 special tokens: one per speaker, two chamber markers, a beginning-of-sequence token, and a presiding officer token. Using `encode(..., allowed_special="all")` rather than `encode_ordinary()` is critical: the latter fragments speaker tokens into meaningless subwords.

B. Additional Results

B.1 Speaker Identification

As a validation exercise, we test whether the model can identify speakers from debate context alone. At each speaker-token position, we restrict predictions to the 1,701 speaker tokens and check accuracy.

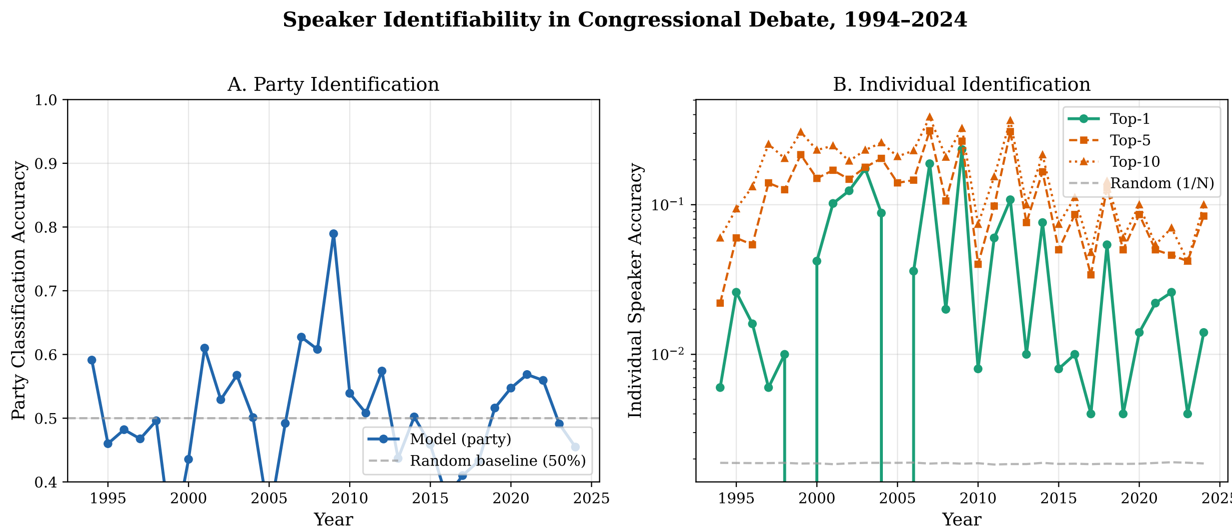

Figure 3: Speaker identification accuracy over time (1994–2024). Panel A: party classification. Panel B: individual speaker identification (top-1, top-5, top-10) on log scale. Random baseline for individual identification is 1/1,701 ≈ 0.06%. Results before 2015 are in-sample.

Party-level accuracy averages 50.6%—near the 50% coin-flip level and *below* the majority-class baseline of 55%, unsurprising since the model predicts individuals and party is derived indirectly. Individual speaker top-1 accuracy averages 4.8%—80 times the random baseline of 0.06%. Top-5 reaches 12.2% and top-10 reaches 17.1%. The model has learned genuine speaker fingerprints from conversational context.

Table 6: Speaker Identification Accuracy

Metric	Mean	Range	Chance Baseline
Party (R vs. D)	50.6%	30.1–78.9%	50.0% (55% majority)
Individual (top-1)	4.8%	0.2–23.4%	0.06%
Individual (top-5)	12.2%	0.6–31.2%	0.29%
Individual (top-10)	17.1%	1.2–38.6%	0.59%

Notes: Accuracy at speaker-token positions across 1994–2024 (pre-2015 in-sample). Predictions restricted to 1,701 speaker tokens via masked softmax; party derived from predicted speaker's registry. Chance baseline = 1/1,701.

B.2 Neural versus Classical Methods

Figure 4 compares the neural model's party classification to a TF-IDF + SVM baseline.

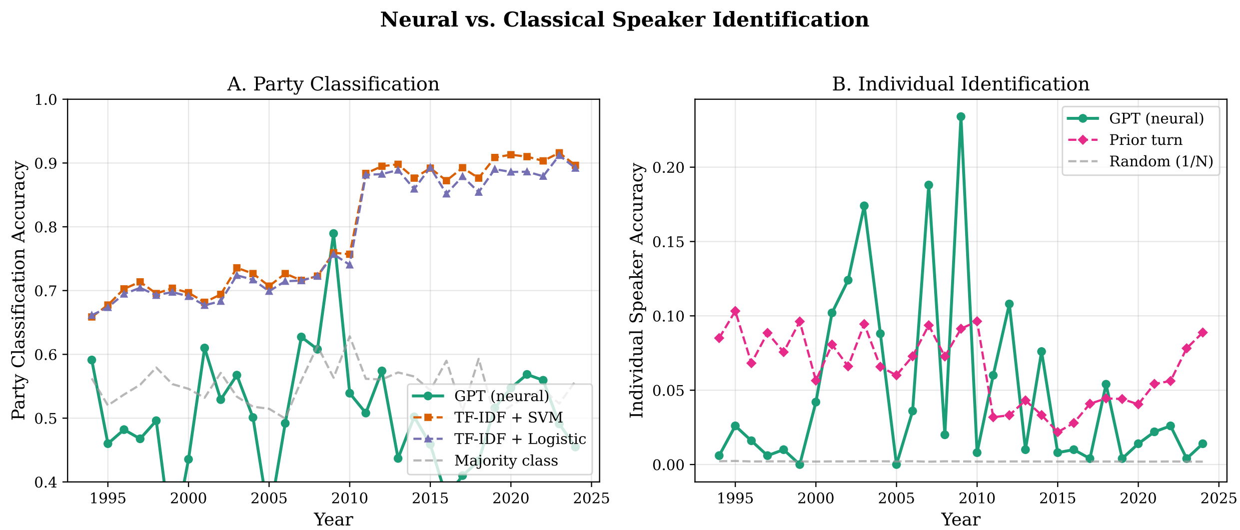

Figure 4: Party classification accuracy: neural model (GPT) versus classical baselines (TF-IDF + SVM, Logistic Regression), 1994–2024. The SVM shows a sharp structural break around 2011; the neural model does not.

The SVM—which classifies party from word frequencies, as in Gentzkow et al. (2019)—shows a dramatic accuracy jump around 2011 (from $\sim$70% to $\sim$90%). The neural model does not. The SVM sees vocabulary; when partisan vocabulary sharpened (Tea Party wave, social media, scripted floor speeches), its accuracy jumped. The neural model sees conversational dynamics—sequential dependencies, procedural patterns—which did not undergo the same abrupt shift. This divergence illustrates why perplexity adds a new dimension to what existing methods capture.

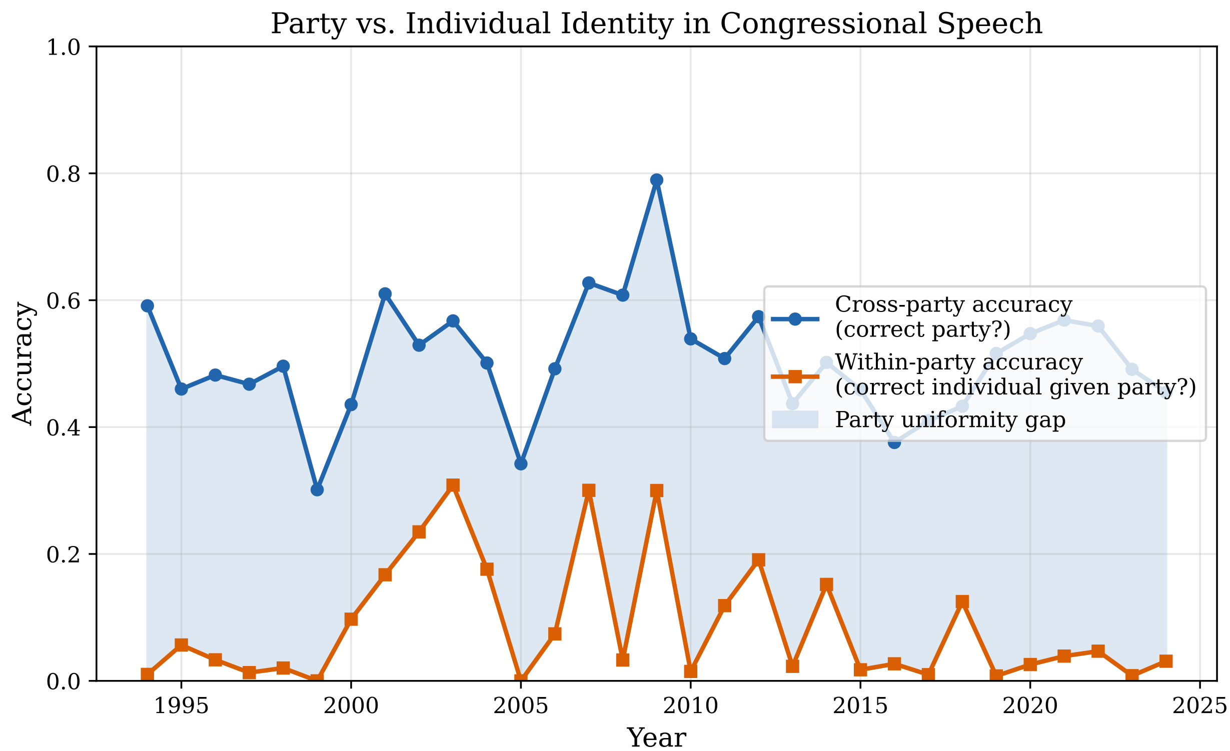

Figure 5: Party identification versus individual identification. Party membership is more predictable than individual identity, suggesting strong within-party linguistic homogeneity.

Table 7: Party Classification Accuracy: All Methods

Method	Mean Accuracy	Range
TF-IDF + SVM	79.4%	65.9–91.6%
Logistic Regression	78.4%	66.1–91.2%
Neural (GPT, restricted softmax)	50.6%	30.1–78.9%
Majority class baseline	55.0%	49.9–62.8%

Notes: Mean and range across 1994–2024. SVM and Logistic Regression use TF-IDF features. Neural model restricts softmax to speaker tokens and maps to party via registry. Pre-2015 in-sample for neural model; SVM retrained per year.

C. Data Pipeline Details

The Congressional Record is downloaded from GovInfo's bulk data API as ZIP archives by date. Each archive contains HTML files for individual debate segments. We parse using the `unitedstates/congressional-record` parser (v2.0.8), extracting speaker BioGuide ID, chamber, text, and turn number. GovInfo rate-limits requests; our script implements exponential backoff.

For HuggingFace data (1994–2010), speaker matching links informal identifiers to BioGuide IDs using the `congress-legislators` dataset (67.7% match rate by speech count, 90.2% by word count). Unmatched speeches are predominantly presiding officer announcements. For GovInfo data, BioGuide IDs are embedded directly; we enriched 618 of 620 unknown speakers from the legislator database.

HuggingFace-era conversations are grouped by (date, chamber); GovInfo-era by document/topic. This difference motivates restricting the Deliberation Index to GovInfo holdout data (2015–2024).